

Greenfield FDI - Internal Methodology for Data Cleaning

Background on Data verification and cleaning task

We utilize FDI data in our economic bulletins to explore the economic relationships between China and various regions. We have three bulletins covering Africa, Latin America, and Asia (new), respectively.

Each bulletin examines different economic trends in China's relationship with other regions, including patterns in development finance, trade, and foreign direct investment. For development finance, we rely on data from GDP Center datasets, including CLA, CODF, etc.. For trade analysis, we use the [UN Comtrade database](#). For FDI insights, we source greenfield FDI data from [FDI Markets](#), which involves thorough verification and cleaning. This internal guidebook explains this verification and cleaning process.

Background on FDI

Foreign Direct Investment (FDI) is when a company or individual from one country invests in a business in another country. There are two main types of FDI:

1. **Greenfield Investment:** This is when a company builds new facilities, like factories or offices, from the ground up in a foreign country.
2. **Mergers and Acquisitions (M&A):** This is when a company buys or merges with an existing foreign company to expand its operations.

Greenfield is like starting fresh, while M&A involves taking over or combining with what's already there.

Foreign Direct Investment (FDI) differs from the lending we track in the Chinese Loans to Africa Database (CLA) or other Global China Initiative (GCI) datasets in several key ways:

- **Nature of Involvement:** FDI involves a private or public institution investing in a company, often in a foreign country, by taking ownership stakes (equity). This means the investor has a lasting interest and control over the business.
- **Types of Borrowers:** Unlike traditional lending, which typically involves loans to state-owned entities or governments, FDI can target private companies in the host country.
- **Balance Sheet Impact:** In lending, the investment appears as debt on the borrower's balance sheet, meaning it must be repaid with interest. In contrast, FDI is recorded as equity, representing ownership in the business. This equity stake can grow or shrink in value depending on the business's success.

In summary, while both FDI and lending involve significant financial commitments, FDI ties the investor more closely to the success or failure of the business through ownership, whereas lending involves a repayment obligation without direct control or ownership.

Introduction to Dataset - FDI Markets

FDI Markets is a data provider specializing in Foreign Direct Investment (FDI) data. They gather data from multiple sources, including government reports, company filings, and press releases.

Their dataset serves as the starting point for our analysis as we undergo a thorough cleaning and adaptation process. One of the key items of this process involves seeing which investments actually went through, removing those that are no longer relevant (zombie projects). There are many cases of large investment pledges made that never materialized, but still appear in the dataset that must be removed during the cleaning process.

How to fill out the Greenfield FDI Dataset:

There are five column blocks in the dataset highlighted in different colors. Each column block contains several associated columns. Some columns are for our own manual input, whereas others contain raw data that is used for reference. Here is a brief overview of the different column blocks, the different columns under them and whether they require a manual input or are immutable (not changeable).

1. **Identifier:** Primary key identifier that identifies an investment project.
 - a. **Columns:** *Deal ID*
 - b. **For Manual Input:** No
2. **Progress:** Contains columns related to reporting work progress and leaving comments and a confidence score about one's findings
 - a. **Columns:** *Reviewer Review Status Analyst Checked Date Confidence Status Duplicate Pair ID (Original) Comments (changes made)*
 - b. **For Manual Input:** Yes
3. **Deal Info:** Contains columns related to the deal information an analyst needs to retrieve as part of the verification process.
 - a. **Columns:** *Deal Status Amount - Announced (Million USD) Amount - Completed (Million USD) CN Deal Stake (%) Date [MM/DD/YYYY] - Announced Date [MM/DD/YYYY] - Completed BU Sector*
 - b. **For Manual Input:** Yes
4. **Sources:**
 - a. **Columns:** *Source (Deal Status) Source (Amount - Announced) Source (Amount - Completed) Source (Deal Stake %) Source (Date [MM/DD/YYYY] - Announced) Source (Date [MM/DD/YYYY] - Completed) Source (BU Sector) Other/General Sources*
 - b. **For Manual Input:** Yes
5. **RAW FDI Markets Data:** These are sourced from the FDI Markets raw data and are immutable in the file so that they cannot be changed and only referenced.
 - a. **Columns:** *Project date FDI Project ID Investing company Parent company Parent company ID Source country Source state Source*

admin-region (US county) Source city Destination country Destination state Destination admin-region (US County) AND MORE

b. For Manual Input: No

6. 2000-2022 Cleaned Data: These columns are sourced from the cleaned file from the previous Bulletin as they occasionally contain 'judgment calls' and associated comments left by former analysts.

a. Columns: Deal Amount used Analyst Comments Project Name Source Used

b. For Manual Input: No

How to fill out the different columns in the dataset Greenfield FDI Dataset:

Column	Description	Manual Input	Categories
Deal ID	Unique primary key to identify project	No	None
Reviewer	Name of designated reviewer	No	None
Review Date	Date when review is finished	Required	None
Analyst	Name of designated reviewer	No	None
Checked Date	Date when project is checked by analyst	Required	None
Confidence Status	The status reflecting the level of confidence in the accuracy of the sources provided. This field is mandatory.	Required	Irrelevant: Projects that do not meet the criteria for inclusion (E.g. it is a loan, aid, an MOU, or it was an investment that simply never materialized) Duplicate: The project is a duplicate entry of another project already in the dataset. Unknown: The available information is insufficient to determine the project's status. Use this when critical details are missing or unclear. High: The project data is highly reliable, with multiple sources verifying 1. Deal Status 2. Amounts 3. Dates

			Medium: The sources of the project data are moderately reliable or only 2/3 fields have information about them: 1. Deal Status 2. Amounts 3. Dates Low: The sources of evidence are of low reliability or only 1/3 fields have information: 1. Deal Status 2. Amounts 3. Dates
Duplicate Pair ID	A reference ID for the original project that is considered a duplicate in the dataset.	Yes	None
Comments	Notes detailing any changes made to the project during the review or verification process.	Yes	None
Deal Status	The current status of the deal, such as announced, completed, or cancelled	Yes	Completed: The project has been fully executed. All activities are finished. Announced: The project has been publicly declared but not yet initiated. Canceled: The project was terminated before completion and will not proceed. Suspended: The project is temporarily halted, with potential to resume later. Commenced: The project has started, but details on progress may vary.

Amount - Announced (Million USD)	The announced investment amount in millions of USD for the project.	Yes	None
Amount - Completed (Million USD)	The actual completed investment amount in millions of USD for the project.	Yes	None
CN Deal Stake (%)	The percentage of the deal that is owned or financed by a Chinese entity.	Yes	None
Date [MM/DD/YYYY]] - Announced	The date when the deal was officially announced.	Yes	None
Date [MM/DD/YYYY]] - Completed	The date when the deal was officially completed.	Yes	None
BU Sector	Based on BU sector options, the industry classification of the project.	Yes	None
Source (Deal Status)	The source of information used to verify the deal status.	Yes	None
Source (Amount - Announced)	The source of information used to verify the announced investment amount.	Yes	None
Source (Amount - Completed)	The source of information used to verify the completed investment amount.	Yes	None
Source (Deal Stake %)	The source of information used to verify the percentage of the deal financed by a Chinese entity.	Yes	None
Source (Date [MM/DD/YYYY]] - Announced)	The source of information used to verify the announcement date of the deal.	Yes	None
Source (Date [MM/DD/YYYY]] - Completed)	The source of information used to verify the completion date of the deal.	Yes	None
Source (BU Sector)	The source of information	Yes	None

	used to verify the business unit sector of the project.		
Other/General Sources	Any additional sources or references that were consulted during the verification and review process.	Yes	None
RAW FDI Markets Data	All columns provided by FDI Markets	No	None
Deal Amount used	The investment amount in millions of USD for the project decided on for the previous edition	No	None
Analyst Comments	Notes detailing any changes made to the project during the review or verification process in the previous edition	No	None
Project Name	The name of the project retrieved for the previous edition	No	None
Source Used	The source used as evidence for the decision made in the previous edition	No	None

QUESTIONS FOR RR:

- Do we have an ownership threshold in mind? e.g. if the greenfield is only 10% owned by a Chinese firm do we include or exclude? What about for M&A?
- If an FDI project has been merely announced but it has not been commenced yet, would you consider that a zombie project that required removal? What about a project that has been announced AND commenced, but not completed yet - would that be removed?
- Does it make sense to do a mapping of sectors so that fdi market sectors map to world bank sectors 1 to 1? My concern is that the world bank focuses on lending whereas these projects are FDI so it might be a bit forced to use WB sectors for different kinds of financing purposes?
- WB Sectors: https://projects.worldbank.org/en/projects-operations/project-sector?lang=en&page=&_gl=1*_aj10dr*_gcl_au*MTYwMDAxODE0NS4xNzE4ODk2NzU5